THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHEMATICS FULL SYLLABUS PAPER SET 10

Full Syllabus (Chapters 1–14) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Solve Q. 29 on graph paper. Draw neat figures.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. If $a = 2^2 \times 3$ and $b = 2 \times 3^2$, then LCM(a, b) is: 1

(a) 36 **(b)** 72 **(c)** 18 **(d)** 6

2. Student A says HCF $\times$ LCM $=$ product holds for any two positive integers; Student B says it holds for any three. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

3. If α, β are the zeroes of $x^2 - 8x + 15$, then $\alpha + \beta + \alpha\beta$ is: 1

(a) 120 **(b)** 7 **(c)** 23 **(d)** −7

4. A student states: “A linear polynomial can have two zeroes.” The correct response is: 1

(a) Correct **(b)** Incorrect — it has exactly one zero

(c) Correct if the constant is 0 **(d)** Cannot be determined

5. The number of solutions of $2x + 3y = 9$ and $4x + 6y = 18$ is: 1

(a) 1 **(b)** 0 **(c)** infinitely many **(d)** 2

6. The pair $kx + 2y = 5$ and $3x + y = 1$ has a unique solution if: 1

(a) $k = 3$ **(b)** $k \neq 3$ **(c)** $k \neq 6$ **(d)** $k = 6$

7. The roots of $x^2 - 7x + 12 = 0$ are: 1

(a) $2, 6$ **(b)** $1, 12$ **(c)** $3, 4$ **(d)** $-3, -4$

8. If $x = 1$ is a root of $kx^2 - 3x + 2 = 0$, then k is: 1

(a) 3 **(b)** −1 **(c)** 2 **(d)** 1

9. The 4th term from the end of the AP $-11, -8, -5, \ldots, 49$ is: 1

(a) 43 **(b)** 46 **(c)** 37 **(d)** 40

10. The number of two-digit numbers divisible by 3 is: 1

(a) 33 **(b)** 30 **(c)** 31 **(d)** 32

11. A student states: “Two congruent triangles are also similar.” The correct response is: 1

(a) Correct **(b)** Incorrect

(c) Correct only for isosceles triangles **(d)** Cannot be determined

12. If $\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$, then $\triangle ABC \sim \triangle DEF$ by: 1

(a) SSS **(b)** ASA **(c)** AA **(d)** SAS

13. $\sec 60^\circ$ equals: 1

(a) 2 **(b)** $\frac{2}{\sqrt{3}}$ **(c)** $\sqrt{3}$ **(d)** $\frac{1}{2}$

14. Tangents at A and B meet at P. If $\angle AOB = 100^\circ$, then $\angle APB$ is: 1

(a) 50° **(b)** 100° **(c)** 80° **(d)** 40°

15. If the area of a sector is $\frac{1}{12}$ of the area of the circle, its central angle is: 1

(a) 60° **(b)** 36° **(c)** 12° **(d)** 30°

16. The angle swept by the minute hand in 20 minutes is: 1

(a) 60° **(b)** 120° **(c)** 100° **(d)** 20°

17. A cone and a cylinder have the same base and height. The ratio of their volumes is: 1

(a) 2 : 3 **(b)** 3 : 1 **(c)** 1 : 3 **(d)** 1 : 2

18. The median of grouped data is given by: 1

(a) $l + \frac{n - cf}{f} \times h$ **(b)** $l + \frac{\frac{n}{2} - cf}{f} \times h$ **(c)** $l + \frac{cf - \frac{n}{2}}{f} \times h$ **(d)** $l - \frac{\frac{n}{2} - cf}{f} \times h$

19. Two students find the mean of the same data by the assumed mean method; A takes $a = 25$, B takes $a = 35$. Then: 1

(a) Only A is correct **(b)** Only B is correct

(c) Both get the same mean **(d)** Neither can get it

20. The probability of an odd prime on a die is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{2}{3}$

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 20–30 words each.

21. Find the HCF and LCM of 17, 23 and 29 by prime factorisation. What do you observe? 2

22. A student claims that $p(x) = x^2 - 4x + 4$ has two distinct zeroes. Justify or refute. 2

23. Find the nature of the roots of $2x^2 - 6x + 3 = 0$. 2

24. E and F are points on sides PQ and PR of $\triangle PQR$. If $PE = 4$ cm, $QE = 4.5$ cm, $PF = 8$ cm and $RF = 9$ cm, state whether $EF \parallel QR$. Justify. 2

25. Find the distance between the points (a, b) and $(-a, -b)$. 2

26. A student says that $\sin^2\theta + \cos^2\theta = 1$ is true only for $\theta = 45^\circ$. Refute the claim. 2

27. A 1.8 m tall man observes the top of a tower at 45° from a point 20 m away. He concludes that the tower is 20 m tall. Justify or refute. 2

28. TP and TQ are tangents from an external point T to a circle with centre O. Show that $\angle POQ + \angle PTQ = 180^\circ$. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each. Solve Q. 29 on graph paper.

29. Draw the graphs of the following pair of linear equations on graph paper. Determine whether the pair is consistent; if so, find the solution from the graph: $x - y = 1$ and $2x - 2y = 2$. 3

30. Find the mean of: 3

Class	0–10	10–20	20–30	30–40	40–50
Frequency	3	7	15	10	5

31. The angles of a triangle are in AP and the smallest angle is 40°. Find the other angles. 3

32. A quadrilateral ABCD circumscribes a circle with $AB = 2x$, $BC = x + 4$, $CD = 9$ and $DA = 7$ cm. Find x and the sides AB and BC. 3

33. Find the area of the minor segment of a circle of radius 5 cm when the chord subtends 90°. (Use $\pi = 3.14$) 3

34. A die is thrown twice. Find the probability that 3
(i) the sum is 7
(ii) the sum is more than 10
(iii) both numbers are the same.

35. A man in a boat rowing away from a 100 m high lighthouse takes 2 minutes to change the angle of elevation of the top of the lighthouse from 60° to 45°. Find the speed of the boat (in m/min). 3

OR

Two points are on opposite sides of a 60 m tower, in line with its foot. The angles of elevation of its top from them are 30° and 60°. Find the distance between the points. 3

36. A student says the median of the data below is 40. Verify. 3

Class	0–20	20–40	40–60	60–80	80–100
Frequency	6	9	10	8	7

OR

Find the mean by the step-deviation method: 3

Class	10–30	30–50	50–70	70–90	90–110
Frequency	8	12	20	6	4

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 100–150 words each.

37. **Case study:** In a school ground ABCD, lines are drawn 1 m apart along AD. 100 flower pots are placed 1 m apart along AD. Niharika runs $\frac{1}{4}$ of AD on the 2nd line and posts a green flag; Preet runs $\frac{1}{5}$ of AD on the 8th line and posts a red flag. 5
(i) Write the positions of the flags.
(ii) Find the distance between the flags.
(iii) Rashmi posts a blue flag exactly halfway between them. Where should she post it?

OR

Case study: In a classroom, four friends are seated at A(3, 4), B(6, 7), C(9, 4) and D(6, 1). Champa says "ABCD is a square", Chameli disagrees. 5
(i) Find AB, BC, CD, DA.
(ii) Find the diagonals AC and BD.
(iii) Who is correct? Justify.

38. If $\tan\theta + \sin\theta = m$ and $\tan\theta - \sin\theta = n$, show that $m^2 - n^2 = 4\sqrt{mn}$. 5

OR

Case study: A wheelchair ramp is in the shape of a right triangle with horizontal base 12 m and vertical rise 5 m. Let θ be the angle of the ramp with the ground. 5
(i) Find the length of the ramp.
(ii) Find $\sin\theta$, $\cos\theta$ and $\tan\theta$.
(iii) Verify $\sin^2\theta + \cos^2\theta = 1$.
(iv) A student says $\tan\theta > 1$ for this ramp. Comment.

39. A water tank is a cylinder with a hemispherical bottom. Its radius is 1.4 m and total depth 3.4 m. Find **5**

(i) its capacity in litres

(ii) the inner surface area to be painted.

OR

A building is in the form of a cylinder surmounted by a hemispherical dome and contains $41\frac{19}{21}$ m³ of air. If the internal diameter of the dome equals the total height of the building above the floor, find the height of the building. **5**

40. The median of the following data is 525 and $\Sigma f = 100$. Find x and y. **5**

Class	0–100	100–200	200–300	300–400	400–500	500–600	600–700	700–800	800–900	900–1000
Frequency	2	5	x	12	17	20	y	9	7	4

OR

Find the median and mean height of the girls: **5**

Height (cm)	120–130	130–140	140–150	150–160	160–170
Girls	2	8	12	20	8

END OF QUESTION PAPER **Total: 80 Marks**

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